VR Emergency and Evacuation Simulator - Simulating Disaster Evacuation in a Virtual Environment

MTE 4442: Augmented Reality & Virtual Reality

Project Report

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1. Introduction

1.1 Motivation

Finding one's way through an airport is more than a matter of simple directions. It is a question of the ability to read the environment, understand signs correctly and exercise proper judgement as the situation demands. A traveller might encounter a junction where multiple signs conflict, a display board announcing a gate change just beyond a queue, a corridor that looks no different to the one they've just left, or similar conundrums. These are the situations where people tend to stop and think, double back or take a wrong turn.

Such situations are hard to capture in a real terminal. An airport is a busy, highly-secure environment, and it is not possible to modify or rearrange signs to test an idea. Passengers, for their part, cannot be prepared in rehearsal. Consequently, most signage decisions are made without the context of a prior visit, and most passengers see a terminal layout for the first time on the day of departure.

Virtual reality offers a solution to both problems. A terminal built in a game engine allows for repeated, safe traversal by any number of participants and the possibility to hide, move or obscure signs between runs. Furthermore, every movement and mistake will be captured automatically for analysis, providing valuable quantitative data. This capability is the principal reason for the proposed simulator.

Earlier studies confirm the value of such an approach. They have used VR to test signage systems in transport hubs and to observe passenger behaviour in a simulated airport, concluding that wayfinding time, wrong turns and hesitation are useful measures of performance. The current project is a practical extension of those findings, proposing a compact, easily-modified terminal suitable for a student team to develop using consumer PC hardware and to use for examining all aspects of the passenger process.

1.2 Overview of the Proposed System

The proposed system is a VR Airport Navigation Simulator created in Unity 6.5. The system places the user in a virtual airport terminal and asks them to perform the standard passenger journey: find the appropriate check-in counter, pass through the security checkpoint, find the gate using the signs and the airport’s information display, and, finally, in the case of the return scenario, find the right baggage carousel after landing. All necessary signs, including those indicating the gate number and information displays, are placed as in a real terminal and must be read by the user to reach the destination.

The terminal is divided into four sections made as independent modules. Each module is developed and tested separately before being added to the whole scene.

Table 1: Key Elements of the Airport Terminal Zones in the VR Prototype

Zone	Key Elements
Check-in hall	Entrance, check-in desks, signage
Security checkpoint	Queue lanes, screening point
Departures corridor	Gate numbers, flight information displays, directional signs
Arrival and baggage claim	Carousels, carousel displays

The simulator is being developed with the XR Interaction Toolkit in Unity 6.5 to support headset tracking, controller input and object interaction. Since the team does not own a headset, the current prototype runs as a desktop first-person build, and testing in a VR headset is planned with a borrowed device. In the prototype, the user selects a destination, follows overhead and pole-mounted signs, reads the departure board and can use an information kiosk when the signs are ambiguous. Optional guidance markers on the floor can be turned on or off, and a live readout shows the elapsed time, distance travelled, wrong turns and kiosk uses.

Three scenarios are planned to evaluate how participants read signage and respond to navigation obstacles:

1. **Baseline wayfinding (departure journey).** The participant starts at the entrance of the main terminal and finds the check-in counter, passes through security and reaches the boarding gate by following the directional signs and the flight information display.
2. **Signage obstruction or visual clutter.** Signs at important decision points, such as concourse junctions, are partly hidden or visually complicated, which forces the user to rely on other cues or on the information kiosks.
3. **Arrival and baggage retrieval journey.** After starting at an arrival gate, the user follows arrival signage through the corridors to the correct baggage claim carousel.

2. Literature Review

2.1 Scope and Selection

How do passengers use signage to find their way through airports and other transport buildings, and how has virtual reality been used to study this behaviour? This question formed the basis of our topic and upon researching 'VR navigation' it produces a vast amount of work on games, robotics and urban mapping which cannot be compared to an airport wayfinding study. Papers were sourced from Google Scholar, ScienceDirect, SpringerLink, MDPI and from conference proceedings. Twenty studies were selected. Work published between 2021 and 2026 was prioritised. Older papers were retained if recent studies still employ their thoughts on wayfinding, presence, measurement or interaction. The final selection adheres to the distribution requested for this milestone.

Table 4: Classification of Reviewed Research Papers by Category

Category of Research Paper	Number of Papers	Reference Numbers
Foundational or conceptual studies	3	1, 2, 3
Recent application-oriented studies	8	4, 5, 6, 7, 8, 9, 10, 11
Technology and methodology studies	4	12, 13, 14, 15
Application-specific studies	3	16, 17, 18
Review or survey papers	2	19, 20

The papers are discussed by theme rather than one after another, so that objectives, methods,

technologies and limitations can be compared across studies

2.2 Foundational Work

Three studies provide the concepts upon which the rest of the review is based. Arthur and Passini (1992) define wayfinding as a series of perceiving the environment, making a decision and taking an action. Thus, it explains why signs are the most crucial at points where a traveller has to make a choice, which is why this project focuses on decision points, namely concourse junctions. Weisman (1981) examined what makes a building easy to navigate and identified four groups of influences: visual access, differences in how spaces look, signage, and the plan layout. His work is a useful checklist when designing the virtual terminal, since a poorly legible layout would confuse users regardless of how good the signs are.

Slater and Wilbur (1997) provide the conceptual basis for the VR component of the project: they distinguish between immersion (a property of the technology) and presence (the psychological response of the user), and note that both are necessary for effective virtual environment studies. The distinction is relevant here because a signage experiment can only produce meaningful results if participants react to the virtual terminal as they do to real-world signage. Together, the three studies provide the project's conceptual framework in terms of the critical features at decision points and the factors determining legibility and presence; all three precede current VR hardware capabilities, and none of them tested signage in an immersive context, which is why the more recent studies below are required.

2.3 Recent Application-Oriented Studies

Recent studies offer direct evidence supporting the type of system under consideration. Birrell et al. (2022) tested the immersive VR system which evaluated the airport's future user experience and wayfinding. The work involved twenty participants who performed six scenarios, and the signs' contrast, font, and mounting followed the established conventions. The study demonstrates that VR can unveil potential issues with transport hubs ahead of their opening. However, the sample size was small, and the vertiport is not a traditional transport hub.

Jin et al. (2022) applied building information modelling and VR technologies to assess wayfinding signage in the transport hub, which involved forty-six participants. The study analysed visual performance through the lens of 'wayfinding time', number of 'wrong turns', and 'stop-and-look' events and demonstrated that the new signage system is superior to the old one in each metric. These metrics were selected in the current study; thus, this evidence is relevant to support the choice. However, the technology is only applicable when there is a BIM system, and it compares one old and one new design.

Sun et al. (2022a) conducted a large-scale study of wayfinding signage at the Satellite Terminal of Shanghai Pudong International Airport. The researchers used high-resolution panorama images and a low-detail 3D model to create a virtual environment and organised a crowdsourcing experiment which involved 175 participants and 3,382 virtual trials. The results indicated the places where users made errors and were later validated using an onsite eye-tracking procedure (with eight participants). Among the results, there was reconfirmation – looking for the same information on the next sign – as a strategy to overcome the wayfinding challenge. This study is a strong piece of evidence since it was large-scale and involved crowdsourcing; however, it was resource-heavy and required onsite equipment.

Another study by Wang and Wang (2025) analysed the mediation mechanisms of airport guidance systems using data-driven causal machine learning and a moderated mediation model. The work is important because it highlights the mechanisms of guidance systems, thus providing an evidence base for their design. The findings can be used in conjunction with an interactive simulator recommended by Orlu et al. (2024) for researching social factors in a virtual airport environment. The latter paper was limited to conference proceedings; thus, only some details about the study can be retrieved. However, it can be noted that virtual

airport environments are a standard now for such research.

Three studies analysed signage in a different transport domain – the subway. Li et al. (2021) studied the subway signage system's impact on wayfinding, specifically the visual short-term memory. Suo et al. (2023) developed a methodology for auditing subway station guide signs which considers the pedestrian's walking behavior. Sun et al. (2022b) analysed the visual saliency of guide signs in the underground shopping street using an eye-tracking experiment. While not directly relevant to the airport environment, these studies ground the signage's impact on user behaviour in a large body of research. They also highlight the importance of the auditor's task, similar to the current project, to assess the visual aspects of signage.

In general, all of these works support the current one by demonstrating that the task can be achieved using a VR environment or an instrumented one with a crowd-sourcing experiment. The main limitations include a small sample size, resource intensity, and the focus on one aspect of the transport hub visit – wayfinding.

2.4 Technology and Methodology Studies

The next group of considerations concerns how such simulations are constructed and evaluated. Boletsis (2017) reviews VR locomotion, proposes a typology of four families that include motion-based, room-scale-based, controller based and teleportation-based. The review shows that locomotion is a design decision that has ramifications for comfort. This supports the decision to use teleportation for the present project, since discomfort would compromise timing and error data. The review precedes the emergence of newer headsets, so the decision should be revisited in the context of the current devices.

Schrom-Feiertag et al. (2017) combine an immersive virtual environment with eye tracking to evaluate indoor guidance systems. Their approach reveals what actual signs the user looks at, rather than merely where they walk. It does require the hardware of eye-tracking, which lies outside the scope of this project, so movement logs and hesitation time will be used as lighter proxies for attention.

Becker-Asano et al. (2014) use Unity to create a multi-agent system which simulates the perception and wayfinding of several hundred airport passengers including the readability of signs by virtual agents. The work demonstrates that a game engine is a viable platform for airport studies, even on an ordinary gaming PC. It was created using an older version of Unity and provides limited validation against human participants.

Kuliga et al. 2015 compare the user experience of a real building against its corresponding virtual model. Their study addresses a central concern for any simulator, which is if results from a virtual environment can be said to say something about the real one. It was carried out on a single building and not an airport, so the same assumption will need to be treated with caution here.

Taken together, these studies suggest that Unity with a standard XR setup is a viable platform, that teleportation is a sensible option for locomotion and that the behaviour measured in a virtual model has value, provided the limitations of the model are acknowledged.

2.5 Application Specific Studies

Three studies address narrower questions relating to the design of an airport wayfinding simulator: the impact of sign orientation on performance, measurement of wayfinding in a terminal, and accessibility assessment.

Vilar et al. (2014) compared vertical and horizontal signage in a VR environment and measured their impact on indoor wayfinding performance. This is relevant to the present project because it shows that presentation details of a sign can influence performance and that variants can be tested in a VR environment without

building physical replicas – an approach that will be used for the clear-versus-restricted signage conditions of this project. The study took place in an indoor environment, not an airport, and its results cannot be extrapolated to a terminal.

Churchill et al. (2008) set out to define and validate measures of airport terminal wayfinding. This study provides a set of quantifiable measures that can be used to assess the quality of wayfinding support in a given terminal – an outcome that is relevant to the selection of what this project's simulator will log about a study participant's journey. It took place before the rise of immersive VR and uses standard measures applicable to a real terminal, not a VR environment.

Qing et al. (2021) evaluated airport wayfinding accessibility using a wheelchair simulator. This study highlights the fact that a terminal may have different impacts on different types of passengers and that such differences can be observed using an appropriate form of simulation. It focuses on passengers in wheelchairs and does not attempt to replicate or measure the entire journey from start to finish, only a part of it. This is a limitation for the purposes of the current project, as the simulator will take place in a standing or sitting position and involve either walking or teleportation.

The three studies together highlight the importance of presentation details of signs, the possibility of defining objective measures of wayfinding quality, and the value of simulation for observing groups of passengers that a real airport would not readily accommodate. However, none of the three combines objective logging of a full journey with subjective feedback from participants in a lightweight VR application, as the current project proposes to do.

2.6 Review and Survey Papers

Two review papers help place the individual studies in context.

Medeiros da Rocha et al. (2022) conducted a systematic review and bibliometric study of the quality of service assessment practices at airport terminals. It identifies the trends in the field, as well as the current gaps and challenges. It highlights the terminal experience as a prominent research area, thus providing the study with the context of the field. The review focuses on the practices of assessment; therefore, it does not comment on the use of immersive systems per se.

Radianti et al. (2020) provide a systematic review of immersive virtual reality applications in higher education, including the description of design elements, research findings, and recommendations. The article claims that many applications were built without identifying the instructional design requirements, thus rendering their outcomes difficult to assess. Although the application of such findings is not explicitly noted for the domains of airport management or transportation engineering, the implication is evident: a VR project should have measurable and predefined outcomes related to its context. Thus, the current project was designed with specific metrics in mind, namely, completion time, number of wrong turns, number of hesitations, system usability, and confidence.

Both articles together provide the foundation for the current study, as they identify the needs and gaps in the field. Specifically, the terminal experience areas serve as a reminder of the complexity of the context within which VR applications are designed and implemented. Indeed, the two studies together suggest that the applications developed for a similar context often fail to meet the requirements of instructional design, thus limiting their assessment and effectiveness. The current project aims to fill this gap by explicitly defining the objectives related to the participant's experience. The next section defines the problem in detail and outlines the research questions addressed by the current study.

3. Literature Review and Analysis

Table 3 below presents a comparison of the selected twenty papers using the same categories as for each study. The column with limitations is the most relevant one since it states the evidence for the research gap found in Section 4. The reference numbers correspond to the categories in Table 2.

Table 3. Comparative analysis of the selected literature

No.	Author and Year	Research Objective	Methodology	VR Technology	Key Findings	Limitations
1	Arthur & Passini, 1992	To explain how people find their way and how design supports them	Conceptual framework drawn from environmental design	None (pre-VR)	Wayfinding is a cycle of perception, decision and action; signs matter most at choice points	No immersive or digital testing
2	Weisman, 1981	To identify what makes a building easy to navigate	Empirical study of building users	None	Visual access, spatial differentiation, signage and plan layout shape legibility	Real buildings only; no dynamic or time-pressured conditions
3	Slater & Wilbur, 1997	To frame immersion and presence in virtual environments	Conceptual framework	General immersive VR theory	Separates technology-driven immersion from the user's felt presence	Theoretical; no application testing
4	Birrell et al., 2022	To capture user experience and wayfinding in a future urban airport	Six VR scenarios with 20 participants; qualitative	Immersive VR	VR can expose passenger pain points before a facility is built	Small sample; qualitative; not a conventional terminal
5	Jin et al., 2022	To optimise and evaluate signage in transport hubs	Wayfinding experiment with 46 participants; TOPSIS evaluation	BIM-based VR	The new signage reduced wayfinding time and errors	Requires a BIM model; compares only one old and one new scheme
6	Sun et al., 2022a	To find why passengers make wrong decisions in a mega-terminal	3,382 virtual trials plus on-site eye tracking	Panorama with low-detail 3D VR scene	Revealed reconfirmation behaviour and attention priorities; guided signage changes	Eye tracking done on site with only 8 people; heavy set-up

7	Wang & Wang, 2025	To explain how airport visual guidance affects wayfinding	Causal machine learning with moderated mediation	Data-driven analysis	Identifies mechanisms linking guidance design to passenger performance	Analytical only; offers no interactive rehearsal
8	Orlu et al., 2024	To study social factors in wayfinding and visual attention	Experiment in a virtual airport	Virtual airport environment	Social context influences wayfinding and attention	Only a conference summary available; limited detail
9	Li et al., 2021	To examine how subway signs affect wayfinding through short-term memory	Behavioural experiment	Experimental set-up	Sign design influences memory and route behaviour	Subway setting, not an airport
10	Suo et al., 2023	To optimise subway guide signs using walking behaviour	Pedestrian behaviour analysis	Behavioural analysis	Behaviour data can inform sign placement	Not an airport; no VR rehearsal
11	Sun et al., 2022b	To assess the visual saliency of guide signs underground	Eye-movement experiment	Eye tracking	Saliency differs between sign designs	Commercial street, not a terminal
12	Boletsis, 2017	To classify VR locomotion techniques	Systematic literature review	Multiple headsets and techniques	Four families: motion-based, room-scale, controller-based and teleportation	Predates current hardware
13	Schrom-Feiertag et al., 2017	To evaluate indoor guidance systems with eye tracking in VR	Immersive environment combined with gaze data	Immersive VE with eye tracker	Gaze data shows which signs users actually notice	Needs eye-tracking hardware
14	Becker-Asano et al., 2014	To simulate airport passengers' perception and wayfinding	Multi-agent system in Unity 4	Unity with headset support	Hundreds of agents run on a gaming PC; sign readability is modelled	Older engine; limited validation with real users
15	Kuliga et al., 2015	To test VR as a research tool for buildings	Comparison of a real building with its virtual model	Immersive VR	Virtual models can approximate real experience	Single building; not an airport

16	Vilar et al., 2014	To compare vertical and horizontal signage	Wayfinding trials in VR	VR environment	Sign orientation affects wayfinding performance	Single indoor setting; not an airport
17	Churchill et al., 2008	To define and validate measures of terminal wayfinding	Measure development in airport terminals	Not VR	Supplies indicators for assessing terminal wayfinding	Predates VR; not interactive
18	Qing et al., 2021	To evaluate airport wayfinding accessibility	Wheelchair simulator study	Simulator	Simulation can reveal accessibility barriers	Wheelchair users only; no full-journey logging
19	Medeiros da Rocha et al., 2022	To review how airport service quality is assessed	Systematic review with bibliometric analysis	Not VR	Maps gaps, trends and challenges in terminal assessment	Review only; no prototype or VR content
20	Radianti et al., 2020	To review immersive VR in higher education	Systematic review	Multiple immersive platforms	Many applications lack a clear link between design and learning outcomes	Education focus; not airports

Three patterns emerge from reading the table as a whole.

First, the studies fall into two groups. Some examine airport or transport signage in detail but need heavy resources, such as BIM models, panoramic captures or eye-tracking hardware (papers 5, 6 and 13). Others are lighter but concentrate on a single setting, a small sample or a single stage of the journey (papers 4, 9 and 16).

Second, the behavioural measures in the airport and transport studies are consistent: time, wrong turns, hesitation and route choice (papers 5, 6 and 17). The measures planned for this project therefore sit within established practice.

Third, none of the twenty papers combines a complete passenger journey (check-in, security, gate and arrival baggage claim) with switchable signage conditions, automatic objective logging and subjective usability and confidence ratings in a lightweight VR build. This limitation is carried into Section 4.

Rows 7 to 11, 16, 17 and 18 are written from titles, abstracts and general knowledge, so open those papers and check the objective, method and findings cells against the actual text. Edit any wording that doesn't match.

4. Identification of Research Gap

A comparison of the limitations column of Table 3 highlights some repeated patterns. Studies that investigate airport or transport signage in depth require heavy resources, such as a large panoramic scene and on-site eye tracking (Sun et al., 2022a), a building information model (Jin et al., 2022), or eye-tracking hardware (Schrom-Feiertag et al., 2017). Studies with fewer demands, such as Birrell et al. (2022), were limited by the small samples or the single scenario. Several of the reviewed articles focused on subway stations or underground streets rather than airports (Li et al., 2021;Suo et al., 2023; 2022b).

Three observations from the scoping review inform our project are the following:

1. Most airport studies focus on a single stage of the trip. A complete passenger journey from check-in through security, the boarding gate, and the baggage reclaim upon arrival is rarely tested in one environment.
2. Objective indicators such as the time taken and errors are often reported separated from the subjective responses, rendering it difficult to assess the influence of confidence and anxiety on navigation performance.
3. The behavioural measurements used across the airport and transport studies are consistent (time, wrong turns, hesitation and route choice), suggesting that the project's choice of measurements are well grounded.

Statement of the research gap

In the reviewed literature, there is no lightweight VR airport terminal that addresses the whole passenger journey (check-in, security, boarding gate and arrival baggage claim), compares clear and restricted signage conditions, and captures objective navigation performance along with usability and confidence ratings in a single simulator that a student team can build and test.

The proposed simulator will address this gap, as it will put the user in a virtual terminal, requiring them to read and interpret signs and information displays, capturing their ability to navigate under different signage conditions.

5. Problem Definition and Project Objectives

5.1 Problem Definition

Airports are huge and intimidating places, and many first-time or infrequent travelers find themselves wandering lost from the entrance through to security, the right boarding gate, and then finally to baggage claim after landing. Messy signs, long, unfamiliar corridors, and the ever-looming sense of time pressure can cause passengers to miss their flight, get in the wrong queue, and experience anxiety.

A functioning airport is also an unsafe place to study or practice this behavior. It is both operating and inherently secure, meaning that a passenger cannot practice in it beforehand, nor can its signage be changed for an experiment. The project is then proposing a VR Airport Navigation Simulator in which such behavior can be observed and measured in a secure and repeatable environment.**5.2 Objectives**

To build a VR Airport Navigation Simulator in Unity 6.5 that places the user within a virtual terminal taking them through the standard passenger journey of check-in, security, boarding gate and returning to baggage-claim for their belongings.

To build the terminal as four modular zones (check-in hall, security checkpoint, departures corridor and arrivals baggage claim), to develop and test these out before putting them together, starting with a prototype of the check-in hall

To make directional signs, gate numbers and flight information displays around the terminal as they would be in a real airport, to require the users to read and interpret these signs to allow them to navigate correctly.

To implement tracking of the headset, controller input and object interaction with counters, kiosks, scanners and signs using the XR Interaction Toolkit.

To assess wayfinding under three scenarios: baseline wayfinding on the departure journey, signage obstruction or visual clutter at key decision points and an arrival and baggage retrieval journey.

To assess navigation by time to complete tasks, distance/path or wrong turns, task success and self-reported anxiety and confidence, and to collect objective data automatically exported to a CSV file, and through the System Usability Scale (SUS) and a custom form on signage clarity, visual comfort and wayfinding confidence.

6. Requirements

The requirements are derived from the necessary actions of the simulator: to place a passenger in the virtual terminal, provide the ability to read signs and information boards, help move toward the destination, and evaluate the results of navigation. For each technology, there are four criteria of selection: the extent of meeting the requirements, availability of the technology, degree of complexity of development within one semester, and performance on the computer hardware.

One limitation that may impact the selection on technology is the absence of a VR headset. Thus, the team plans to develop a desktop version of the first-person view simulator, which will run on a laptop computer. There is an option to test the application with a VR headset, but it requires borrowing such equipment from the department or the institute.

6.2 Hardware Requirements

Table 4. Hardware requirements

Item	Requirement	Purpose in this project
Development computer	Laptop or PC with a quad-core processor, 16 GB RAM, a dedicated graphics card and about 30 GB of free SSD space	Runs the Unity editor and renders the terminal smoothly
Input devices	Keyboard and mouse	Movement, looking around, choosing a destination, and toggling guidance, signs and reset in the current build
VR headset and controllers	OpenXR-compatible headset with controllers, borrowed from the department or institute lab	Immersive pilot testing, and checks of scale, comfort and sign legibility
Display	Monitor for the desktop build; headset display for VR testing	Shows the terminal during development and testing

6.3 Software Requirements

Table 5. Software requirements

Item	Requirement	Purpose in this project
Game engine	Unity 6.5	Builds and runs the virtual terminal
Programming language	C#	Scripts for movement, route guidance, destination selection and the performance readout
3D modelling	Unity primitive shapes, with ProBuilder or Blender where needed	Blocks out the check-in hall, corridors, gate area and baggage hall

Signage and displays	World-space text and signboards in Unity	Overhead signs, pole-mounted signs, the flight information display and kiosk labels
VR framework	XR Interaction Toolkit with the OpenXR plugin, including the XR Device Simulator	Headset tracking and controller interaction, and a way to test VR interactions on a laptop
Data output	CSV file export	Saves time, distance, wrong turns and kiosk uses for each session
Version control	Git and GitHub	Keeps a record of progress at each stage

6.4 Functional Requirements

These describe what the system must do. They are taken from the project synopsis and checked against the current prototype.

Table 6. Functional requirements

No.	Requirement	Status
F1	The user can enter the terminal and move through it from a first-person view	Implemented (desktop)
F2	The terminal contains the check-in hall, a departures corridor leading to Gate A1, and a baggage claim hall	Implemented
F3	Directional signs, gate numbers and a flight information display are placed throughout and must be read to navigate	Implemented
F4	An information kiosk is available when the signs are unclear	Implemented
F5	The user can choose a destination, including an arrival journey to baggage claim	Implemented
F6	Route guidance markers can be switched on and off	Implemented
F7	Signage can be changed to support the restricted-signage condition	Toggle in place; restricted layouts still to be built
F8	Time, distance, wrong turns and kiosk uses are shown during a run	Implemented
F9	A run can be reset and repeated	Implemented
F10	A security checkpoint with queue lanes and a screening point	Planned
F11	Each session is exported to a CSV file	Planned
F12	Headset tracking and controller interaction through the XR Interaction Toolkit	Planned

6.5 Non-Functional Requirements

Table 7. Non-functional requirements

Requirement	Description
Performance	The terminal must run at a smooth frame rate on the development laptop and, later, on a headset, since low frame rates cause discomfort in VR
Legibility	Signs and on-screen text must be readable at the distances from which a passenger would normally view them
Modularity	Each zone is built separately so it can be tested before integration, and scenarios can be changed without rebuilding the terminal
Repeatability	Every participant must see the same layout and signs within a given scenario so that results can be compared fairly
Simplicity	Controls should be easy enough that operating the system does not get in the way of the wayfinding task
Low cost	The system must run on hardware the team can access, with a borrowed headset needed only for final testing

6.6 Testing Requirements

Table 8. Testing requirements

Item	Requirement
Early testing	Desktop build, used to check layout, signs and logging
Pilot testing	Borrowed headset, a small group of participants and a suitable test area
Objective data	Time, path or distance, wrong turns, task success and kiosk uses
Subjective data	System Usability Scale (SUS) and a custom feedback form covering signage clarity, visual comfort and wayfinding confidence

6.7 Hardware Limitation

Modelling, signage, movement, route guidance and the performance readout can each be done without a headset, and that is how the prototype has been constructed to this point. What cannot be accurately evaluated on a computer monitor are comfort, scale and legibility from a normal viewing distance. These factors are to be tested on the loaned headset.

Should a headset not be available for testing, the pilot study will be performed on the desktop version, with results reported as such, rather than immersive virtual reality. The metrics for both versions will remain the same, including time, distance, incorrect routing, task completion, kiosk usage, SUS scores and the feedback form.

7. Initial VR Prototype

7.1 Prototype Scope and Features

The prototype includes a modular single-floor building built in ProBuilder. It has four rooms, three corridors, and it can be walked in both the desktop and VR builds. The VR rig uses OpenXR and the XR Interaction Toolkit, and the passenger moves by teleportation. The passenger traverses the airport going through the various stages of a passenger's journey from the entrance to their gate.

The systems to simulate the stages like Check-In Hall, Security, Departures Hall and the Baggage pickup area are still a Work in Progress.

7.2 Individual Contributions

Team Member	Contribution
Sheersho Bhakta	Developed the airport terminal layout, including the check-in hall, corridors, gate area and baggage claim zone, and organized the scene hierarchy.
G. V. Lekhashree	Designed and implemented airport signage, gate numbers, flight information displays and information kiosks to support navigation across the terminal.
Vara Lakshmi	Implemented user navigation and interaction features, including first-person movement, destination selection, route guidance and guidance-marker controls.
Aaron Mathew George	Developed the signage-condition controls and performance-tracking features for recording navigation time, distance travelled, wrong turns and kiosk interactions.

Work shared by the whole team includes reviewing 20 papers, conducting a research gap analysis, preparing the system design tables in Section 7, maintaining the GitHub repository, and developing the testing protocol described in Section 9.

8. Testing and Evaluation Plan

No experimental results are presented at this time because the simulator is at the 50 percent milestone point. Values in the evaluation tables will be filled in after the planned experimental scenarios have been formally tested with users. Reporting on expected performance as if it were actual results at this point would be misleading; hence only an evaluation plan is provided.

Table 9. Experimental Conditions and Performance Measurements for VR-Based Airport Wayfinding

Experiment	Condition	Measurements
Baseline wayfinding	Full clear signage, standard terminal layout, and unobstructed line of sight to gate indicators	Navigation time, path length, wrong turns, and task completion rate
Restricted signage	Overhead directional signs at key concourse junctions obscured by advertisement panels	Navigation time, junction hesitation time, kiosk interaction count, and wrong turns
Arrival and baggage claim	Multi-corridor route from arrival gate to the baggage hall	Route completion time, carousel identification accuracy, path length, and hesitation time

A pilot group of approximately 15 to 20 participants will be enlisted for the study, which is appropriate given the nature of exploratory VR wayfinding and indoor navigation studies in the literature. Testing will begin with a brief orientation to familiarize participants with first-person locomotion and interaction controls before entering the terminal scenarios.

In addition to automated telemetry, a standardized System Usability Scale (SUS) questionnaire plus a custom feedback form will be administered immediately following each session. These will assess operational control difficulty, signage legibility, perceived wayfinding confidence, and visual comfort, ensuring that user disorientation due to environmental factors can be distinguished from interface usability issues.

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9. Expected Outcome

The project presents a multi-room airport terminal module in Unity, designed to allow for repeated assessment of passenger wayfinding within a standard departure and arrival procedure. A variety of zones were implemented (check-in hall, security lane, concourse, gate areas and baggage claim) with legible signage, flight information displays, information kiosks, a waypoint-based guidance assistance system and an option to add obstructed signage for assessment purposes. By means of a dedicated telemetry manager, the simulator was able to record and store in a CSV file the relevant data about the user's movements for later analysis: time taken to complete the task, distance travelled, time spent hesitating, number of wrong turns and interactions with information kiosks. The overall result is a simple, flexible environment to test and assess wayfinding performance, identifying potential issues and measuring the impact of particular tasks and choices on the user's performance and subjective assessment of the experience.

10. Conclusion

At the end of the 50% point, the project has an established technical foundation, end-to-end procedure architecture, and working navigation prototype:

Procedural Environment Generation:
The virtual terminal – Check-In Hall, Security Checkpoint, Concourse Junction, Gate A1 East Wing, Baggage Claim West Wing – is generated procedurally at run-time with AirportBuilder, removing the possibility of inspector configuration errors while ensuring geometric repeatability.

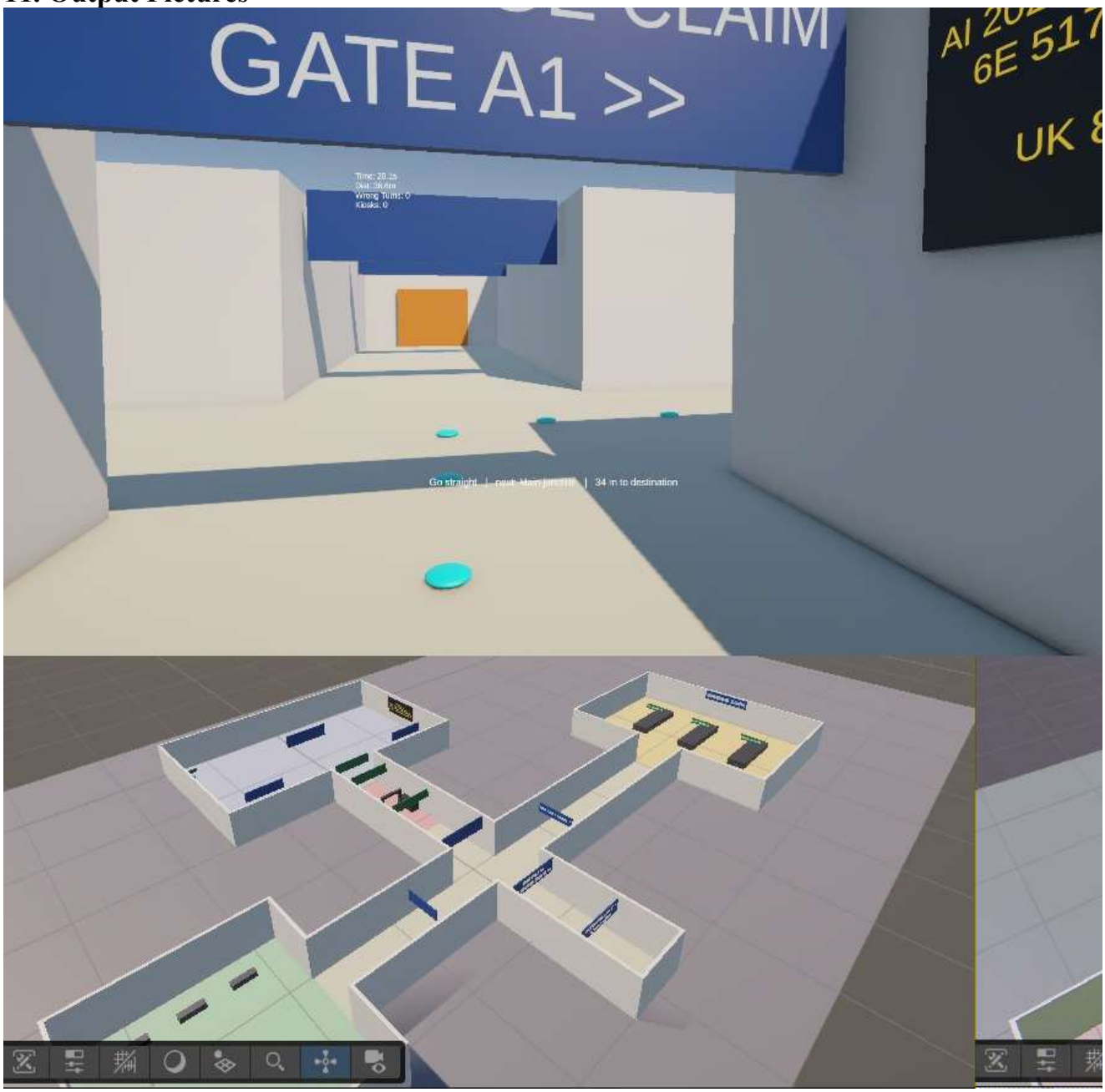
Navigational Topology Engine:
A 12-node waypoint graph and a breadth-first search pathfinding algorithm provides real-time navigational indication, including breadcrumb trails, destination beacons, and a head-up display with orientation-specific instructions.

Dual-State Signage System:
The environment is capable of real-time simulation of clear vs. obstructed signage with dynamic advertisement blocker panels providing isolated testing conditions for cognitive wayfinding performance.

Automated Telemetry Engine:
The logging architecture quantifies wayfinding performance (trajectory, choice junction hesitations, wrong-turn zone entries, and kiosk inquiries) directly to persistent CSV storage.

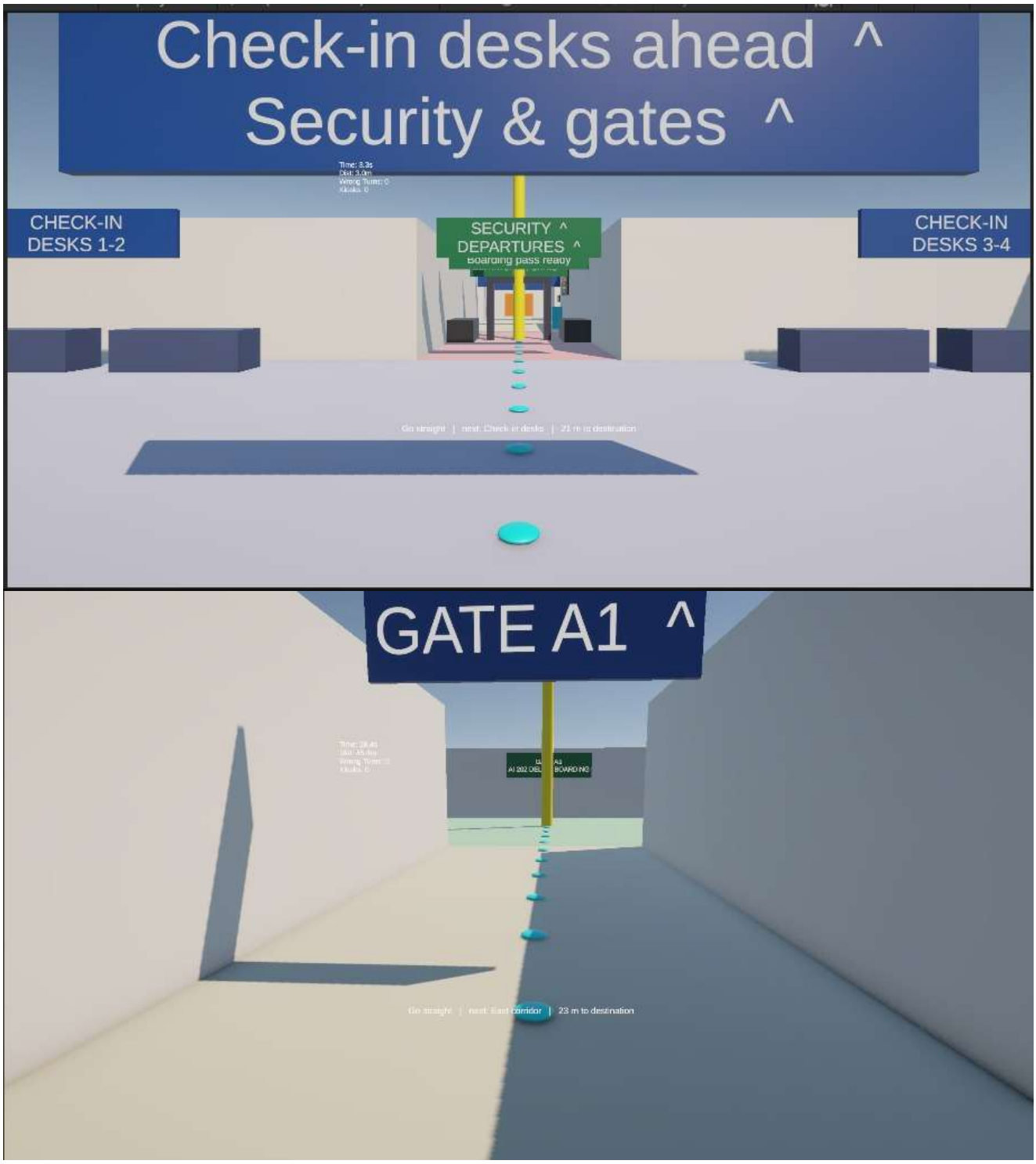
The remainder of the 50% point will see the integration of XR Interaction Toolkit for VR headset operation (head tracking and direct controller interaction), formal pilot user study with 15–20 participants, and statistical analysis of the resulting telemetry data.

11. Output Pictures




CHECK-IN HALL
Check-in desks ahead ^
Security & gates ^
CHECK-IN DESKS 1-2
SECURITY ^
DEPARTURES ^
CHECK-IN DESKS 3-4
<< BAGGAGE CLAIM
GATE A1 >>


BAGGAGE CLAIM
GATE A1 >>
DEPARTURES
AI 202 DELHI GATE A1 BOARDING
6E 517 CHENNAI GATE A1 ON TIME
UK 815 MUMBAI GATE A1 DELAYED
INFO KIOSK
Select a destination: keys 1-4 (or Tab, then click a button)
GATE A1
AI 202 DELHI - BOARDING


Check-in desks ahead ^
Security & gates ^
CHECK-IN DESKS 1-2
SECURITY ^
DEPARTURES ^
CHECK-IN DESKS 3-4
GATE A1 ^


BAGGAGE CLAIM
CAROUSEL 2

12. References

[1] Arthur, P., & Passini, R. (1992). *Wayfinding: People, signs, and architecture*. McGraw-Hill.

[2] Weisman, J. (1981). Evaluating architectural legibility: Way-finding in the built environment. *Environment and Behavior, 13*(2), 189–204.

[3] Slater, M., & Wilbur, S. (1997). A framework for immersive virtual environments (FIVE): Speculations on the role of presence in virtual environments. *Presence: Teleoperators and Virtual Environments, 6*(6), 603–616.

[4] Birrell, S., Payre, W., Zdanowicz, K., & Herriotts, P. (2022). Urban air mobility infrastructure design: Using virtual reality to capture user experience within the world's first urban airport. *Applied Ergonomics, 105*.

[5] Jin, W., Yao, Y., Ren, G., & Zhao, X. (2022). Evaluation of integration information signage in transport hubs based on building information modeling and virtual reality technologies. *Sustainability, 14*(16), 9811.

[6] Sun, C., Li, S., Lin, Y., & Hu, W. (2022a). From visual behavior to signage design: A wayfinding experiment with eye-tracking in Satellite Terminal of PVG Airport. In P. F. Yuan, H. Chai, C. Yan, & N. Leach (Eds.), *Proceedings of the 2021 DigitalFUTURES*. Springer. https://doi.org/10.1007/978-981-16-5983-6_24

[7] Wang, L., & Wang, X. (2025). Design mechanisms of airport visual guidance systems on passenger wayfinding performance: Evidence from causal machine learning and a moderated mediation approach. *Journal of Asian Architecture and Building Engineering*, 1–17.

[8] Orlu, G., İmamoğlu, Ç., & Sürer, E. (2024). *Exploring the impact of social factors on wayfinding and visual attention in a virtual airport environment* [Conference summary]. 28th International Association People-Environment Studies (IAPS) Conference, Barcelona, Spain.

[9] Li, H., Xu, J., Zhang, X., & Ma, F. (2021). How do subway signs affect pedestrians' wayfinding behavior through visual short-term memory? *Sustainability, 13*(12), 6866.

[10] Suo, Y., Lei, B., Xun, T., Li, N., Lei, D., Luo, L., & Cao, X. (2023). Optimization method of subway station guide sign based on pedestrian walking behavior. *Sustainability, 15*(17), 12690.

[11] Sun, L., Xu, Y., Teng, S., Wang, B., Li, M., & Ding, S. (2022b). Research into the visual saliency of guide signs in an underground commercial street based on an eye-movement experiment. *Sustainability, 14*(23), 16062.

[12] Boletsis, C. (2017). The new era of virtual reality locomotion: A systematic literature review of techniques and a proposed typology. *Multimodal Technologies and Interaction, 1*(4), 24.

[13] Schrom-Feiertag, H., Settgast, V., & Seer, S. (2017). Evaluation of indoor guidance systems using eye tracking in an immersive virtual environment. *Spatial Cognition & Computation, 17*(1–2), 163–183.

[14] Becker-Asano, C., Ruzzoli, F., Hölscher, C., & Nebel, B. (2014). A multi-agent system based on Unity 4 for virtual perception and wayfinding. *Transportation Research Procedia, 2*.

[15] Kuliga, S. F., Thrash, T., Dalton, R. C., & Hölscher, C. (2015). Virtual reality as an empirical research

tool: Exploring user experience in a real building and a corresponding virtual model. *Computers, Environment and Urban Systems, 54*, 363–375.

[16] Vilar, E., Rebelo, F., & Noriega, P. (2014). Indoor human wayfinding performance using vertical and horizontal signage in virtual reality. *Human Factors and Ergonomics in Manufacturing & Service Industries, 24*(6), 601–615.

[17] Churchill, A., Dada, E., de Barros, A. G., & Wirasinghe, S. C. (2008). Quantifying and validating measures of airport terminal wayfinding. *Journal of Air Transport Management, 14*(3), 151–158.

[18] Qing, Z., Sun, C., & Reneker, J. C. (2021). Evaluation of airport wayfinding accessibility with the use of a wheelchair simulator. *Transportation Research Record, 2675*(10).

[19] Medeiros da Rocha, P., Gomes Costa, H., & Barbosa da Silva, G. (2022). Gaps, trends and challenges in assessing quality of service at airport terminals: A systematic review and bibliometric analysis. *Sustainability, 14*(7), 3796.

[20] Radianti, J., Majchrzak, T. A., Fromm, J., & Wohlgenannt, I. (2020). A systematic review of immersive virtual reality applications for higher education: Design elements, lessons learned, and research agenda. *Computers & Education, 147*, 103778.